Acquirers Overpay, Markets Learn Slowly: A Market-Neutral 8-K Event Strategy Found by a Pre-Registered, Multiple-Testing-Controlled Search

Gator Quant Hacks 2026 · Systematic Trading Track · Team: [team member names] · Code: [public GitHub URL] · Reproduce: `python event8k/replay.py` (Table 1, no key) · full rebuild: `event8k/build_grid.py` → `grid.py --oos` → `acq_analysis.py`

1. Summary

When a US company worth $2B or more files an 8-K announcing an **acquisition agreement** (Massive 8-K disclosure category), we short its stock at the next close, hedge market beta with SPY, and hold for 60 trading days, at 2% of capital per event. The rule was selected *blindly* from a pre-registered grid of 165 event × holding × size configurations, using in-sample data only (Jan 2022–Sep 2025). Its single out-of-sample run (Oct 2025–Sep 2026) confirmed it. Net of 10 bps per side and a 1%/yr borrow fee: **in-sample Sharpe 0.98** (8.3%/yr, max drawdown −7.6%), **out-of-sample Sharpe 0.82** (9.9%/yr, −12.2%), with beta to SPY of −0.02. Every calendar year from 2022 to 2026 is positive.

> **What we claim, and what we do not.** Over the full 4.7 years, t = 2.01; shorting the same stocks on random dates never beat it (0 of 100 placebos); and it survives doubled costs plus a 3% borrow fee. But it was chosen from 165 cells and does **not** survive Benjamini-Hochberg-Yekutieli or Holm correction in-sample. Part of its return is the 2022–26 large-cap premium (SMB loading −0.49; factor-adjusted alpha +3.2%/yr, t = 0.93). It is the most statistically supported of the 32 hypotheses and 165 grid cells we tested, all reported, and it agrees with a 40-year literature on acquirer underperformance. It is not proven alpha.

2. Economic Hypothesis

Why acquirers drift down. Bidders systematically overpay. Management hubris and the winner's curse in competitive auctions (Roll 1986), empire building, and optimistic synergy forecasts mean that the average large deal destroys acquirer value (Moeller, Schlingemann & Stulz 2005). The market reacts to the announcement, but only partly: acquirers keep underperforming for months to years afterwards (Loughran & Vijh 1997; Savor & Lu 2009 for stock-financed deals), as integration costs, financing and diluted guidance arrive gradually. **Who is on the other side:** holders anchored to management's synergy story, analysts who revise slowly, and index and passive owners who keep the stock regardless. **Why it persists:** each deal is idiosyncratic and hard to value in real time, shorting is costly and squeeze-prone, and the information arrives as soft text (8-K exhibits), not numbers. **Testable predictions** (checked in Section 5): the drift comes from acquirers, not deal targets; it is event-specific (not a property of these stocks on other dates); it builds over weeks, not on day 1; it is not a market-beta bet.

How we found it. After 32 pre-registered hypotheses (Section 5.4), we committed a grid to git *before* running it (`event8k/GRID_PLAN.md`): 34 event definitions × {20, 60, 120} sessions × {≥$2B, ≥$10B}, with a fixed winner rule (best BHY survivor, else highest deflated Sharpe).

None survived BHY, and this cell was the fallback winner. The rationale explains a pattern found by search, so the out-of-sample test is the evidence that matters.

3. Data & Universe

Events: Massive Filings & Disclosures API (sponsor data), 8-K category `acquisition_agreement`, all US filers, 2022-01-01 to 2026-09-30 (taxonomy coverage begins in 2022), with one event per filer per 30 days. **Universe:** filer market cap ≥ $2B *on the day before filing* (Massive point-in-time reference data), giving 1,109 events across 705 tickers. Firms later acquired or delisted remain in the sample; the 3.7% of events with no point-in-time cap are excluded. **Prices:** Massive daily bars, split- and dividend-adjusted; SPY is the hedge. **Access:** Massive REST API with the hackathon key: `/stocks/filings/8-K/vX/disclosures` (events), `/v3/reference/tickers/{t}?date=` (point-in-time cap), `/v2/aggs` (bars). About 60k requests are cached on disk; the raw data is licensed and not committed, and `event8k/build_grid.py` rebuilds it. **Attribution only:** Kenneth French 5 factors + momentum; CBOE VIX. **Split:** 4.75 years of data, so the track's 20% rule binds (0.95 years < 2 years). In-sample is 2022-01-01 to 2025-09-30; out-of-sample is 2025-10-01 to 2026-09-30, run once for the winner. (An earlier, different 8-K strategy was evaluated once on this window; the winner was chosen without it.) **Missing data:** never forward-filled from later data; a stock that stops printing is carried at its last close.

4. Methodology

- **Timing:** 8-K filing dates carry no time of day and can be after the close, so we enter at the close of the first session *strictly after* the filing date. No same-day trading.
- **Position:** short 2% of capital per event; long SPY × β (β from sessions −260 to −11, clipped to [0.5, 1.5]); hold 60 sessions, exit at the close. Gross short is capped at 100% of capital: new events are skipped while the book is full (13% of IS entries, 33% OOS).
- **Costs:** 10 bps per side on stocks (half-spread plus slippage for ≥$2B names) and 2 bps on SPY; borrow fee 1%/yr on short notional (general collateral for liquid large and mid-caps). Stress cases: 2× costs and 3% borrow. Sharpe is computed on excess returns, with T-bill interest on cash and the short rebate excluded.
- **Selection statistics:** two-sided p-values on in-sample Sharpe t-stats across all 165 valid cells; BHY FDR control (valid under dependence); Holm; deflated Sharpe (Bailey & López de Prado 2014) with N = 165. The direction (long/short) was set by the in-sample sign, which the two-sided tests account for.
- **Daily process (as it would run live):** (1) after the close, pull new `acquisition_agreement` 8-Ks; (2) keep filers with prior-day market cap ≥ $2B; (3) estimate β from past returns; (4) send market-on-close orders next session (short stock 2%, buy β × 2% SPY), skipping entries if gross short would exceed 100%; (5) close the stock and its hedge at the 60th close. The hedge is set at entry and not rebalanced in between, so the backtest P&L includes β drift.
- **Out-of-sample protocol:** grid, selection rule and trade rules were committed before any run; the winner was chosen on in-sample data; the OOS year was run once (`grid.py --oos`) and reported as is.

- **Evidence beyond Sharpe:** random-date placebo, P&L decomposition, execution lags, neighbouring cells, FF5 + momentum regression (Newey-West).

5. Results

Net of costs and borrow	Ann. return	Vol	Sharpe	Max DD	Turnover	Worst month
In-sample Jan 2022–Sep 2025	8.3%	8.5%	**0.98** (t 1.90)	−7.6%	14.9×/yr	−3.5%
Out-of-sample Oct 2025–Sep 2026	9.9%	12.1%	**0.82** (t 0.82)	−12.2%	16.4×/yr	−6.9%

Table 1. Required metrics, in-sample and out-of-sample reported separately, every number net of 10 bps/side, 2 bps on SPY and 1%/yr borrow.

Stress test (Sharpe, net)	In-sample	Out-of-sample
Base case	0.98	0.82
2× trading costs	0.88	0.74
2× trading costs + 3%/yr borrow	0.68	0.58
Enter 1 day later / 5 days later	0.92 / 1.12	0.77 / 0.77

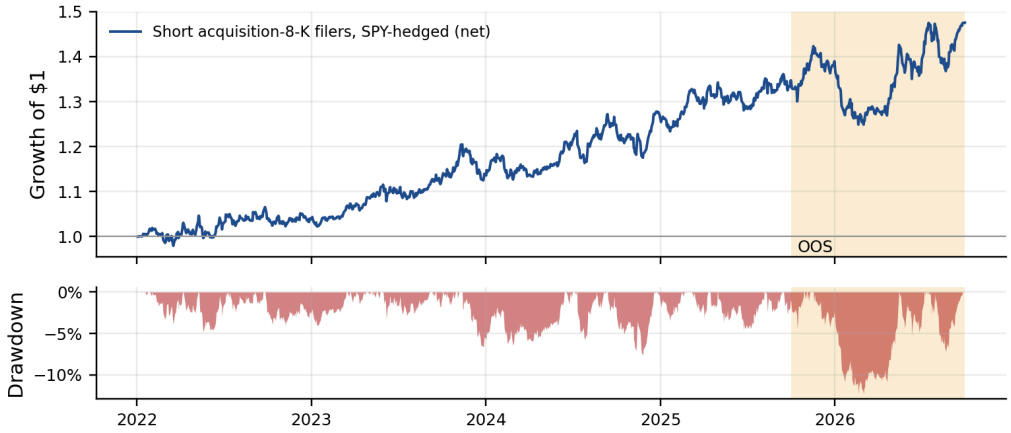


Figure 1. Growth of $1 net of costs and borrow, and drawdown. Shaded: the out-of-sample year, evaluated once. Calendar years: 2022 +3.8%, 2023 +9.8%, 2024 +11.0%, 2025 +9.9%, 2026 YTD +6.2%.

5.1 Is it the event, or something else?

- **Random-date placebo:** shorting the same stocks (behind the 732 in-sample events) on random dates (≥90 days from their deals), with identical sizing and hedging, gives mean Sharpe **−0.53** (95th percentile −0.13). **0 of 100** placebos reach 0.98. On ordinary dates these stocks beat their hedge; after an acquisition filing they lag it.
- **Acquirers, not targets:** per event over 60 sessions, the stock returns −0.07% while its β-hedge earns +2.61%. Likely deal targets (post-filing volatility below half the pre-filing level, i.e. pinned to the deal price) are only 43 events (+0.85%); the other 689, mostly acquirers, earn **+2.79%** each. This is the predicted source.

- **Not beta, not timing:** β to SPY is −0.01 IS / −0.14 OOS; P&L on ±2% SPY days is about 0. Entering 1 or 5 days late barely matters (0.92, 1.12): a slow drift, no lookahead, and room to execute.
- **Neighbours (IS Sharpe):** ≥$2B: 20d 0.62, **60d 0.98**, 120d 0.88; ≥$10B: 0.36, 0.41, 0.83. All six are positive (a plateau, not a spike); the effect is weaker in mega-caps, which are more efficiently priced.
- **Factor attribution (full period):** market β ≈ 0.01, **SMB −0.49**, R^2 0.44, alpha +3.2%/yr (t = 0.93). Short mid/large caps against cap-weighted SPY earns the 2022–26 large-cap premium. The placebo has the same size tilt and loses, so the post-filing window adds to it, but we cannot reject zero factor-adjusted alpha.

5.2 Multiple testing, stated plainly, and what to expect live

In-sample, 165 cells had at least 40 events; none survives BHY at 5% or Holm; the winner's p = 0.058 and deflated Sharpe probability 0.67. The runners-up are also shorts after bad or complex 8-K news (control weaknesses, joint CEO+CFO exits, restatements), consistent with slow incorporation but individually insignificant. The out-of-sample year (0.82, 303 new events) is what lifts this above a search artefact; one year is still short (95% CI on OOS Sharpe ≈ ±2). **Expected live Sharpe: about 0.5** (≈ 4–5%/yr at 8–10% vol). That is the IS/OOS average of about 0.9, haircut by roughly half for selection from 165 cells (deflated Sharpe probability 0.67) and for the size-factor share of returns.

5.3 Strategy 2: cross-asset momentum (diversifier)

Our first, fully pre-registered strategy (rules committed before any data, commit `6b6764d`): 12-month time-series momentum on 10 ETFs, with volatility-scaled positions, monthly rebalancing and a VIX overlay. Net of costs and financing: IS 2008–Sep 2024 Sharpe 0.29, OOS Oct 2024–Sep 2026 Sharpe 0.72; correlation with AQR's TSMOM factor 0.72; **correlation with the acquirer short −0.10**. It is a credible but low-Sharpe trend premium (in-sample t ≈ 1.2) whose OOS gain was mostly equity beta (0.50). Details are in Appendix B and `strategy/`.

5.4 What did not work: 32 hypotheses + a 165-cell grid, all reported

Research line (pre-registered in git)	Trials	Outcome
Pilots: 8-K → volume/direction, Gemini earnings text, weather vs Kalshi	4	all failed
Momentum variants + 10-idea ETF screen + portfolio	16	2 near-misses failed OOS; momentum = Strategy 2
Themes: rainfall → softs, GLP-1 news, open-source AI	3	no effect
Massive options (bonus note)	9	no tradable edge
8-K stocks: abrupt CFO exits (0.52 / 0.32 OOS); 165-cell grid	4 + 165	this strategy

6. Risk Management

- **Limits (implemented):** 2% of capital per name, so a single name can lose at most about 2% × its 60-day rise. Gross short ≤ 100%, with the SPY hedge sized to β; net market

exposure is about zero by construction. On average 43 names are held (mean gross 86% IS, 97% OOS), which is well diversified across sectors.

- **Squeeze risk (the main tail):** the worst positions were IonQ (+83% in 60 days, −1.7% of capital), Rocket Lab (+68%, −1.2%) and QXO (+59%, −1.1%): speculative high-momentum acquirers. Daily VaR95 is 0.97%, ES99 1.78%, worst day −2.4%, skew −0.16, excess kurtosis 0.7: no hidden short-volatility profile.
- **Regimes:** Sharpe 0.65 when VIX > 25 vs 0.97 otherwise; every calendar year positive; ex-best-year (2024) Sharpe 0.86. The OOS drawdown (−12.2%, early 2026) came in a risk-on rally in speculative stocks.
- **Factor exposure:** SMB −0.49. A size-neutral hedge (an equal-weight or mid-cap ETF instead of SPY) is the obvious refinement, but it would be a new trial and is untested here.
- **Proposed live rules (untested, so new trials):** stop out a name at +40% against us; skip top-decile 12-month-momentum names (squeeze-prone); halve gross after a −10% drawdown; drop names with borrow above 3%.

7. Liquidity & Capacity

Entries have a median 20-day dollar volume of $72M (10th percentile $17M). With 2% positions and square-root impact ($\sigma_{daily} \cdot \sqrt{(trade/ADV)}$ per side, on top of the fixed 10 bps):

Strategy AUM	$10M	$50M	$100M	$250M	$500M
Median / 95th-pct trade as % of ADV	0.3% / 1.6%	1.4% / 7.7%	2.8% / 15%	6.9% / 39%	14% / 77%
Extra impact drag (%/yr)	1.0	2.3	3.3	5.2	7.3

Against an 8–10% net return, **impact takes a third of the edge near $100M and half near $250M**, so the realistic capacity is about $100–150M. Larger books should cap trades at 5% of ADV and work entries over 2–3 days (a 5-day delay costs nothing). There are about 190–200 entries a year (about 4 trades a week). ≥$2B shorts are generally easy to borrow, and the 3%-borrow stress keeps Sharpe at 0.58–0.68. A prime/margin account carries the short book plus the hedge (about 1.7× gross at full deployment). At full deployment the 100% gross cap skipped 13% (IS) to 33% (OOS) of entries; a 1.5% position size would admit them all at the same gross.

8. Limitations & Next Steps

- **Selection from a search:** no cell survived multiple-testing correction; the evidence rests on one OOS year, the placebo and the literature. A second OOS year is the real test. Our live positions are hash-stamped in `FORWARD_LEDGER*.md`.
- **Size factor:** factor-adjusted alpha is not significant. Next: a size-matched hedge and a split between cash and stock deals (Savor & Lu predict stock-financed deals drive the effect).
- **Short history:** 8-K categories exist only from 2022. Extending with SEC EDGAR Item 1.01/2.01 parsing back to 2004 would allow decades of testing.
- **Classification:** acquirer vs target is inferred from volatility, and Massive tags are model-assigned. Hand-checking a sample is next.

References

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Data: Massive (8-K disclosures, reference data, daily bars, options); Yahoo Finance; CBOE VIX; Kenneth French Data Library; AQR data sets; US Treasury; Federal Reserve; FRED; Open-Meteo. Tools: Python, pandas, NumPy, SciPy, matplotlib.

Appendix A (outside the page limit). 8-K grid, in-sample, top cells by p-value (all 165 in `event8k/output/grid_is.csv`)

Event · size · hold	n	Dir	Sharpe	t	Halves	BHY q	DSR
acquisition_agreement · ≥$2B · 60	732	short	0.98	1.90	1.17 / 0.70	1.00	0.67
internal_control_weakness · ≥$2B · 20	79	short	0.92	1.77	0.56 / 1.24	1.00	0.62
acquisition_agreement · ≥$2B · 120	732	short	0.88	1.69	0.69 / 0.93	1.00	0.59
CEO_AND_CFO exits · ≥$2B · 60	75	short	0.88	1.69	0.64 / 1.12	1.00	0.59
financial_restatement · ≥$2B · 20	43	short	0.87	1.68	0.84 / 0.98	1.00	0.59
acquisition_agreement · ≥$10B · 120	285	short	0.83	1.60	0.89 / 0.73	1.00	0.56
divestiture_agreement · ≥$10B · 20	153	short	0.83	1.60	1.66 / 0.07	1.00	0.56
auditor_dismissal · ≥$2B · 20	86	short	0.80	1.55	0.93 / 0.66	1.00	0.54

Appendix B. Strategy 2, cross-asset momentum (net of costs and institutional financing)

	Ann. return	Sharpe	Max DD	Notes
In-sample 2008–Sep 2024	2.8%	0.29	−27.5%	AQR TSMOM corr 0.72; 3/6/9/12m plateau 0.18–0.30
Out-of-sample Oct 2024–Sep 2026	8.7%	0.72	−13.3%	market β 0.50; alpha t −0.07
Stress: GFC / 2022 / COVID	+16.8% / +23.5% / −3.2% (SPY −46% / −24% / −33%); loses in V-shaped reversals (Feb 2018 −10.9%)			

Appendix C. Nine-pitfall audit

Pitfall	How we addressed it
P-hacking	Every test pre-registered in git; 32 hypotheses + 165 grid cells reported; BHY/Holm/DSR; nothing claimed as significant after correction
Overfitting	No fitted parameters; the grid choices (hold, size) are reported as a plateau; CSCV PBO 0.21 for the ETF screen
Lookahead	Entry the session after the filing date; point-in-time market cap; β from past data; lag 1/5 tests
Survivorship	All filers, delisted included; point-in-time size filter; 3.7% missing caps excluded
Costs	10 bps/side + 1% borrow; 2× costs + 3% borrow still Sharpe 0.68 IS / 0.58 OOS
Test-set leakage	Winner chosen on IS by a committed rule; OOS run once (git log)
Misleading Sharpe	t-stats and CIs reported; no short-vol profile (skew -0.16)
Regime dependence	Every year 2022–26 positive; VIX>25 Sharpe 0.65; size-factor exposure disclosed
Capacity	ADV-based square-root impact; about \$100–150M